

Ibrahim Elsharkawy

ML Researcher & Physicist

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Research interests. Interpretability, AI for Science, uncertainty quantification, and physics-inspired ML theory. My work covers two directions: *Physics for AI* (physics-inspired theory/data for AI interpretability) and *AI for Physics* (generative/foundation models as tools for scientific discovery).

EDUCATION

University of Toronto & Vector Institute

Ph.D. in Physics

Toronto, ON | 2025 - Present

Y. Kahn, C. Madison, D. Curtin

University of Illinois Urbana-Champaign

M.Sc. in Physics

Urbana, IL | 2023 - 2025

Y. Kahn, B. Hooberman

Rice University

B.S. Computational Physics | B.A. Applied Math & Philosophy

Houston, TX | 2019 - 2023

K. Ecklund

RESEARCH

Interpretability of Generative Models for Physical Systems

Toronto, ON | Jan 2026 - Present

University of Toronto & Vector Institute

Doctoral Researcher

- *Physics for AI*: Building generative models that are **interpretable by construction** and constrained to satisfy physical symmetries and conservation laws. We find that **physical data is the unique testbed** by which interpretability methods can be developed and verified [P2].
- *AI for Physics*: Using the same interpretability tools validated on simulated systems **to extract novel physics from generative models** trained on real data.

Foundation Models for Scientific Point Clouds

Berkeley, CA | Mar 2025 - Present

Lawrence Berkeley National Laboratory, NERSC

Doctoral Researcher

- Building **cross-domain** foundation models for scientific point clouds (small irregular graphs) spanning high-energy physics, cosmology, and molecular dynamics [A2] [A4]. We find that pre-training in a domain **with cheap simulation** (e.g. High Energy Physics Proton Collisions) **benefits domains with limited or expensive data** (e.g., molecular dynamics, proteins) due potentially to shared physics.
- Researching optimal pre-training objectives for scientific foundation models when accurate simulators can produce **labeled data at scale**. Is self-supervision superior for downstream tasks when not necessary? [P3].

Neural Uncertainty Quantification

Urbana, IL | 2023 - 2025

University of Illinois Urbana Champaign

Doctoral Researcher

- Derived and empirically validated an **effective-field-theory description of neural networks** that predicts initialization-driven ensemble variance from first principles **without training an ensemble** [P1].
- Developed a contrastive-normalizing-flow methodology for **uncertainty-aware parameter estimation** robust to systematic sources of error. **Won 1st place** in the Higgs Uncertainty Challenge at NeurIPS 2024 and CERN IML [A1] [A3].

INDUSTRY

AI for (Geo)Physics Research Intern

Houston, TX | 2024 - 2026

ExxonMobil

Generative Models with Physical Priors for Imaging Large 3D Volumes with Seismic.

- Researched and developed **"Physical" Flow Matching for Imaging Large 3D Volumes with Seismic** and general **ML parameter-estimation methods for seismic inversion**, targeting the inversion of traditionally degenerate earth properties.
- Built novel ML-based **anomaly detection tools for 4D seismic data**.

Research and Software Engineering Intern

Houston, TX | 2018 - 2023

Petroleum GeoServices

Machine Learning and Cloud Infrastructure for Seismic

- Researched ML methods for seismic first-break picking, contributed to cloud migration and microservices.

 PUBLICATIONS Full list of preprints at ibrahimelsharkawy.com.

Physics for AI

- P1. I. Elsharkawy**, et al. *Uncertainty Quantification from Ensemble Variance Scaling Laws in Deep Neural Networks*. **Machine Learning: Science and Technology**, 6(3):035040, 2025.
- P2.** Bogorad, **I. Elsharkawy**, Kahn, et al. *Generative Models on Phase Space*. arXiv:2604.02415, 2026.
- P3. I. Elsharkawy**, J. Birk, et al. *Pre-Training for Simulation-Based Science: A Study on Jet Foundation Model Training Objectives*. arXiv:2606.14870, 2026.

AI for Physics

- A1. I. Elsharkawy** and Y. Kahn. *Contrastive Normalizing Flows for Uncertainty-Aware Parameter Estimation*. arXiv:2505.08709, 2025.
- A2. I. Elsharkawy**, et al. *OmniMol: Transferring Particle Physics Knowledge to Molecular Dynamics with Point-Edge Transformers*. arXiv:2601.10791, 2026.
- A3.** W. Bhimji, et al. *FAIR Universe HiggsML Uncertainty Dataset and Competition*. **NeurIPS 2025 Datasets & Benchmarks Track**.
- A4.** V. Mikuni, **I. Elsharkawy**, et al. *OmniCosmos: Transferring Particle Physics Knowledge across the Cosmos*. arXiv:2512.24422, 2025.

 TALKS & POSTERS

Invited Talk. *1st-Place Competition Milestone: Ensembles and Uncertainty Quantification*. The Challenge of Handling Uncertainties in Fundamental Science, NeurIPS 2024.

Invited Talk. *Contrastive Normalizing Flows for Uncertainty-Aware Parameter Estimation*. 7th Inter-Experimental LHC Machine Learning Workshop, CERN, 2025.

Poster. *Pre-Training for Science: A Study on Foundation Model Training Objectives*. Stanford Center for Decoding the Universe Forum, 2025.

Poster. *FAIR Universe HiggsML Uncertainty Dataset and Competition*. Competition Track, NeurIPS 2025.

Poster. *Uncertainty Quantification from Scaling Laws in Deep Neural Networks*. Machine Learning and the Physical Sciences Workshop, NeurIPS 2024.

 AWARDS

- 2026** NSERC Doctoral Canadian Graduate Research Scholarship (CGRS D)
- 2025** Connaught International Fellowship, University of Toronto
- 2024** 1st Place, Higgs Uncertainty Challenge, NeurIPS 2024
- 2023** UIUC Graduate College 3-Year Fellowship
- 2021** J. and M. Graham Physics Scholar, Rice University
- 2016** Eagle Scout

 TEACHING

TA and Guest Lecturer PHY 2108/2109, *The Physics of Machine Learning*, University of Toronto, 2026.

TA. PHY 252, *Thermal Physics*, University of Toronto, 2025.

TA. PHYS 413, *Optics Laboratory*, UIUC, 2024.

 SKILLS

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| Languages | Python (proficient), C/C++, MATLAB, Fortran, Bash, HTML/CSS, $\LaTeX$ |
| Preferred ML Libraries | PyTorch, Lightning, Weights & Biases, Hugging Face, TensorFlow |
| Methods | Generative models (flow matching, normalizing flows, diffusion), transformers (Point-Edge Transformers, ViTs, LLMs), Fourier Neural Operators, foundation model pre-training, uncertainty quantification and Bayesian inference |
| Compute/Infrastructure | Large Scale Distributed training (running on $\mathcal{O}(500)$ nodes on NERSC Perlmutter), Slurm, CUDA, Docker, Git |